

Khotso Selialia

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About

Ph.D. candidate in Electrical and Computer Engineering at UMass Amherst specializing in machine learning, deep learning, optimization, and large language models (LLMs). My research focuses on trustworthy and fair ML, optimization stability, and contextual reasoning in multi-modal, federated, and cross-lingual systems. I have experience with transformer-based models, LLM fine-tuning (LoRA, QLoRA), federated optimization, and long-context sequence modeling, with exposure to reinforcement learning, graph-based models, and differential privacy through graduate coursework and research.

Education

University of Massachusetts Amherst

PhD in Electrical and Computer Engineering

2021 – 2026 (Exp.)

Thesis: *Towards Trustworthy and Fair Machine Learning*. Advisor: Prof. Fatima Anwar. Interests: Distributed Learning, Optimization, Responsible AI.

Carnegie Mellon University, Africa

MS in Information Technology

2019 – 2021

Thesis: *Compressing Stereopairs for Virtual Reality: An Entropy-Based Study*. Courses: Applied Stochastic Processes, Deep Learning, Probabilistic Graphical Models, Machine Learning, Computer Vision, Advanced NLP.

National University of Lesotho

BSc in Mathematics and Computer Science

2014 – 2018

Research Interests

Machine Learning & Optimization; Large Language Models & Generative AI; Natural Language Processing & Speech; Computer Vision & Robotics (SLAM, visual tracking); Federated Learning & Distributed Optimization; Information & Knowledge Management; Automated Reasoning & Neurosymbolic Methods; Fairness, Security & Privacy in Machine Learning (differential privacy, bias mitigation).

Technical Skills

Programming: Python (Advanced), Java, C/C++, JavaScript, Node.js

ML & Optimization: Supervised, semi-supervised & unsupervised learning; deep learning; predictive modeling; stochastic gradient methods and adaptive optimization; federated optimization; exposure to reinforcement learning and multi-step decision-making from graduate coursework.

NLP & LLMs: Natural language processing and understanding; multilingual and low-resource machine translation; transformer-based models; large language model fine-tuning with LoRA and QLoRA; sequence-to-sequence modeling; long-context handling and positional encodings.

Speech & Sequence Modeling: Experience with phoneme recognition, CTC-style sequence modeling, and RNN/GRU-based speech recognition from graduate projects; analysis of contextual drift in long sequences.

Computer Vision & Robotics: CNN-based visual modeling; exposure to visual SLAM and tracking from research on human-centered XR and IoT; familiarity with robot perception concepts and multi-sensor fusion.

Security, Fairness & Privacy: Fairness-aware learning in federated and distributed settings; bias measurement and mitigation; practical experience with gradient clipping and Gaussian noise injection (differential privacy-style mechanisms) in optimization.

Cloud & MLOps: AWS (EC2, S3), Docker, Jenkins; scalable ML pipelines and experiment orchestration.

Tools: PyTorch (Advanced), TensorFlow, HuggingFace Transformers, Git, Linux (Ubuntu, Fedora), Bash scripting.

Work Experience

University of Massachusetts Amherst

Research Assistant

2021 – Present

- Designed and implemented **federated optimization frameworks** in Python and PyTorch, improving group fairness metrics by 41% across heterogeneous clients via stability-aware training objectives and careful aggregation strategies.
- Investigated **context-amplified bias** in multilingual NMT systems, modeling how tokenization and long-context sequences behave over time and how this impacts attention stability, predictive performance, and robustness on low-resource languages.
- Developed **DCARPE**, a dynamic positional encoding mechanism that adaptively modulates transformer attention based on fragmentation signals, improving long-context translation accuracy by up to +6% ChrF++ while preserving computational efficiency.
- Deployed DCARPE with **LoRA-based parameter-efficient fine-tuning** of transformer and LLM-style models, achieving 4–6% gains in ChrF++, improved cross-lingual generalization with limited sentence-level supervision, and no additional inference-time latency.
- Conducted empirical studies on **optimization stability**, gradient norms, and client heterogeneity in federated learning, including experiments with gradient clipping and noise injection to study trade-offs between performance, fairness, and privacy.

Publications & Research Highlights

1. **Khotso Selialia**, Antoine Nzeyimana, Fatima Anwar. “*Mitigating Tokenization-Induced Distance Distortion in Long-Context Multilingual Machine Translation.*” In Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (**ACL**), 2026.
2. Yasra Chandio, **Khotso Selialia**, Luis Antonio Garcia, Joseph DeGol, and Fatima M. Anwar. “*Lost in Tracking Translation: A Comprehensive Analysis of Visual SLAM in Human-Centered XR and IoT Ecosystems.*” ACM Transactions on Sensor Networks (**TOSN**), 2025.
3. **Khotso Selialia**, Yasra Chandio, Jimi Oke, and Fatima Anwar. “*Mitigating Intersectional Bias in Federated Learning: Beyond Group Fairness.*” **[under review]**
4. **Khotso Selialia**, Yasra Chandio, Jimi Oke, and Fatima Anwar. “*Optimization Stability for Generalization in Differentially Private Federated Learning.*” **[under review]**
5. **Khotso Selialia**, Yasra Chandio, and Fatima M. Anwar. “*Mitigating Group Bias in Federated Learning for Heterogeneous Devices.*” ACM Conference on Fairness, Accountability, and Transparency (**FAccT**), 2024.
6. Yasra Chandio, Momin A. Khan, **Khotso Selialia**, Luis Garcia, Joseph DeGol, and Fatima M. Anwar. “*A Neurosymbolic Approach to Adaptive Feature Extraction in SLAM.*” IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2024.
7. **Khotso Selialia**, Yasra Chandio, and Fatima M. Anwar. “*Federated Learning Biases in Heterogeneous Edge-Devices: A Case-Study.*” ACM SenSys (**SenSys**), 2022.

Volunteer Experience

ACM IMX, ACM SenSys, NeurIPS, AISTATS, ACL, ICML

Conference Reviewer

2025

Reviewed submissions for top-tier conferences, evaluating technical correctness, novelty, and clarity. Provided constructive feedback to authors to enhance the quality and impact of research contributions.

Awards & Honors

2025: College of Engineering Teaching Fellowship, UMass Amherst

2024: ACM FAccT Travel Grant

2019: MINDS Scholar, Mandela Institute for Development Studies

2019: CIT Dean’s Fellowship